MIL-S-9741B
3 March 1960
Superseding
MIL-C-9741A(USAF)
6 May 1958

MILITARY SPECIFICATION

SCALE, WIND, GEOSTROPHIC AND GRADIENT ML-543/UM, ML-544/UM, ML-545/UM (COMPUTER, WIND, GEOSTROPHIC AND GRADIENT ML-543/UM, ML-544/UM, ML-545/UM)

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers the following three types of computers which are used with weather charts to determine geostrophic and gradient winds:

Designation	Chart with which used	Scale of Chart
ML-543/UM	Lambert Conformal Projection	1:10,000,000
ML-544/UM	Polar Stereographic Projection	1:15,000,000
ML-545/UM	Polar Stereographic Projection	1:20,000,000

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal

L-V-351	Vinyl Chloride Polymer and Copolymer Rigid Sheets

Military

MIL-P-116	Preservation, Methods of
MIL-P-2215	Paper; Book, Offset
MIL-N-7513	Nomenclature Assignment, Approval of Nameplate Identification and Descriptions; Aeronautical Support, Ground Electronic, and Photographic Equipment, Contractors Method for Obtaining
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery

FSC 6660

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STANDARDS

Military

MIL-STD-129	Marking For Shipment and Storage
MS-33586	Metals, Definition of Dissimilar

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction.-This specification makes provision for preproduction testing.

3.2 Materials, parts, and processes.-Materials, parts, and processes shall be as specified herein.

3.2.1 Plastic.-The vinyl chloride plastic used in the body and cursor of the computers shall conform to L-V-351, class 1, clear, with a high polish finish.

3.2.2 Paper.-The paper insert (3.5.1.3) shall be white and shall meet the requirements of MIL-P-2215.

3.2.3 Metals.-Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion in atmospheric conditions likely to be met in storage or normal service.

3.2.3.1 Dissimilar metals.-Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

3.3 Design.-Each of the three types of computers shall consist of a body (3.5.1), a cursor (3.5.2), and a swivel assembly (3.5.3).

3.4 Performance requirements.-The components of the computers shall retain the specified degree of flatness (3.6.2), permanency of bonding (3.5.1.4), and transparency (3.5.1.4), when subjected to the following conditions:

3.4.1 Service conditions, operating.-The computers will be used in shelters with temperatures varying from 0 degrees Centigrade (C) to 52 degrees C (32 degrees Fahrenheit (F) to 125 degrees F) and relative humidities varying from 0 to 100 percent.

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3.4.2 Service conditions, storage.-The computers will be stored under temperature conditions ranging from minus 40 degrees C to plus 54 degrees C (-40 degrees F to +130 degrees F) and relative humidities from 0 to 100 percent.

3.5 Details of components

3.5.1 Body.-The body of each computer shall consist of a bottom plastic sheet, a paper insert, and a top plastic sheet.

3.5.1.1 Marking

3.5.1.1.1 Scales.-The bottom of the top plastic sheet shall be marked with the scales of figure 1, 2, or 3, whichever is applicable (3.5.4 and 6.3). The figure also shows the desired relation of the marking to the edges of the top plastic sheet. This marking must be accomplished in reverse so that it can be read from the top of the computer.

3.5.1.1.1.1 Color.-The ten curved lines in the central scale of each computer and the numbers which they intersect shall be green. All other markings shall be black.

3.5.1.1.2 Insert marking.-The top of the paper insert shall be blank so that no marking underlays the scales. The bottom of the insert shall contain the following information printed in capital letters approximately 3/64 inch high:

COMPUTER, WIND GEOSTROPHIC AND GRADIENT
ML-*
STOCK NO.*
(manufacturer's name and address)
CONTRACT NO.*
U.S.

INSTRUCTIONS FOR USE

I. GEOSTROPHIC WIND - USE BOTTOM SECTION OF SCALE (G VERSUS L)

A. TO OBTAIN WIND FROM A "CONTOUR" OR "D" CHART.

1. PLACE POINTERS ON CHART TO SPAN 200 FEET OF CONTOUR HEIGHT OR D VALUE.

2. READ OR INTERPOLATE GEOSTROPHIC WIND (G) SPEED IN KTS AT INTERSECTION OF CURSOR LINE AND PERTINENT LATITUDE (L) LINE.

* fill in appropriate number (3.5.1.1.2.1)

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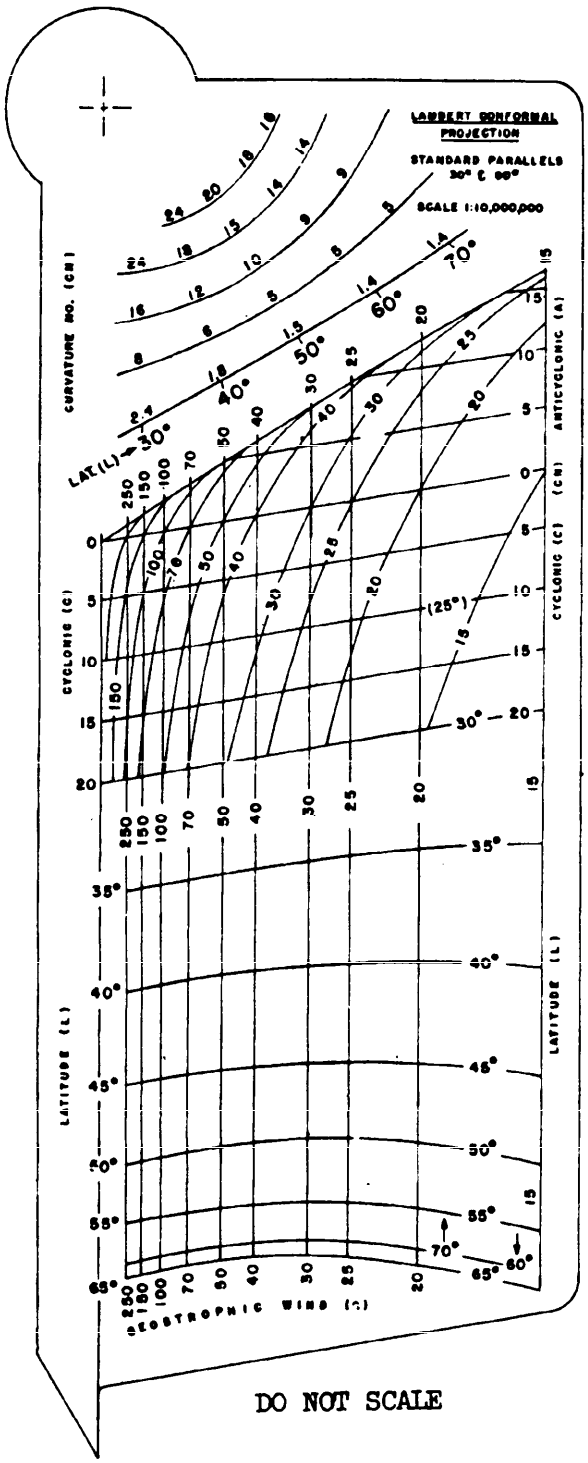


FIGURE 1 SCALE LAYOUT (ML-543/UM)

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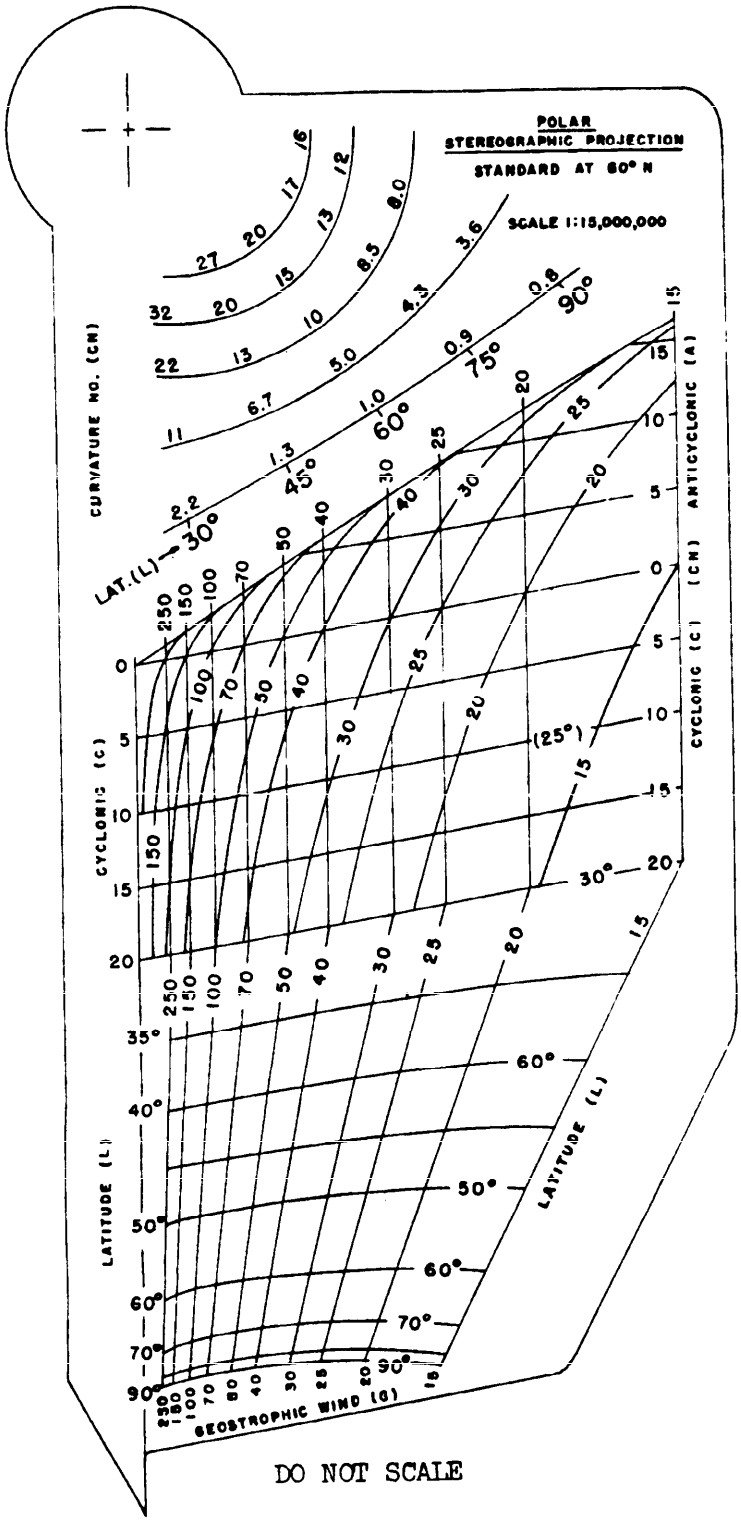


FIGURE 2 SCALE LAYOUT (ML-544/UM)

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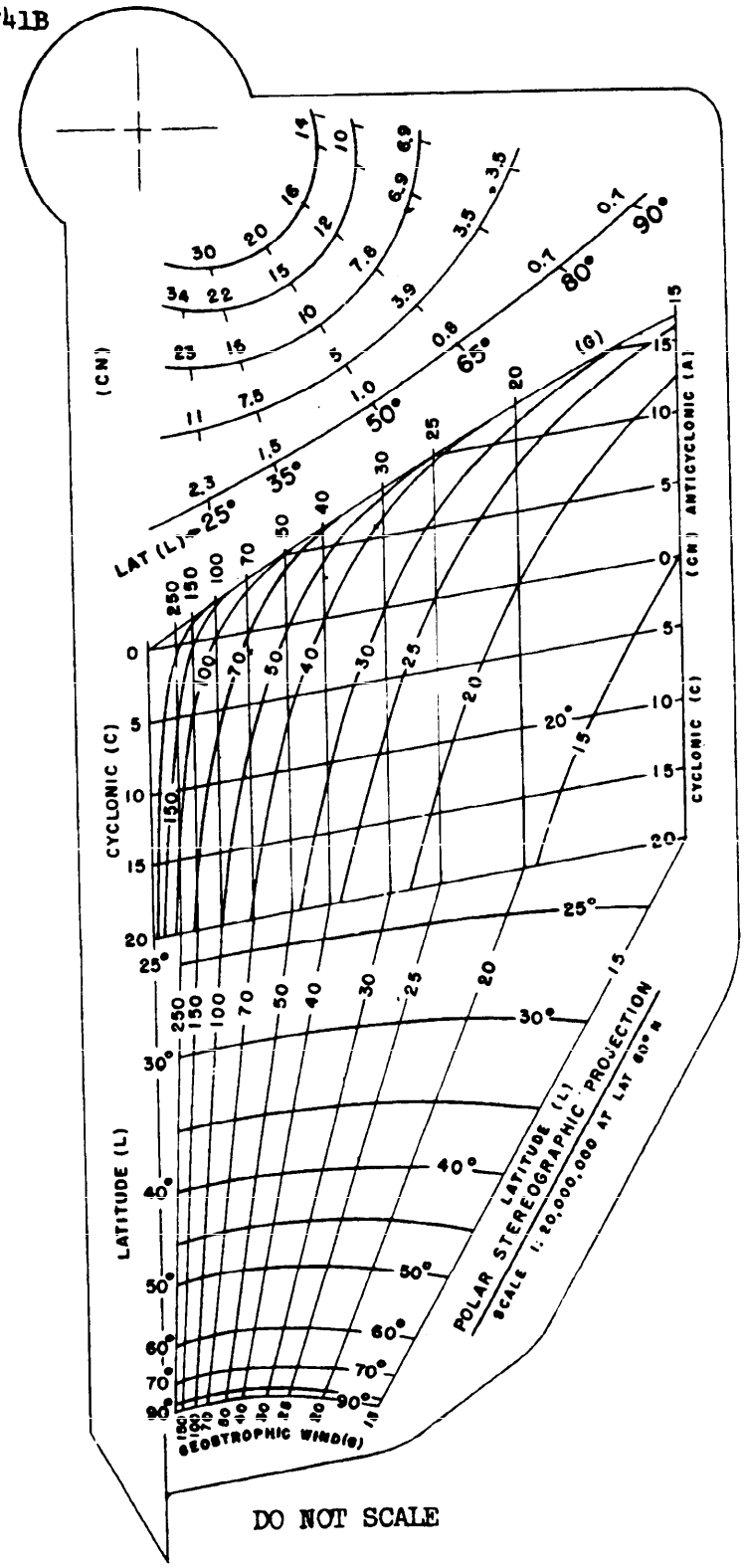


FIGURE 3 SCALE LAYOUT (ML-545/UM)

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3. DIRECTION OF COMPUTED WIND IS PERPENDICULAR TO A LINE BETWEEN THE POINTERS WITH LOWER HEIGHTS TO THE LEFT (NO. HEM.).

B. TO DETERMINE GEOSTROPHIC SPACING OF 200 FT CONTOUR OR D-LINES USE REVERSE PROCEDURE OF A.

II. GRADIENT WIND - USE UPPER AND CENTER SECTIONS OF SCALE (CN AND CN VERSUS G)

1. LAY BLACK CURVED LINES OVER CHART TO MATCH ONE OF CURVED LINES WITH THE CHART CONTOUR LINE.

2. READ OR INTERPOLATE CURVATURE NUMBER (CN) FOR THE PERTINENT LATITUDE.

3. LOCATE, ON CENTER GRAPH INTERSECTION OF CN AND G. (USE LOWER PART OF CENTER GRAPH FOR CYCLONIC CURVATURE (C), UPPER PART FOR ANTI-CYCLONIC CURVATURE (A).

4. READ THE GREEN LINES FOR GRADIENT WIND SPEED (SEE AWSM 105-50/4)

3.5.1.1.2.1 Nomenclature.-Nomenclature assignments and approval of nameplate identification shall be obtained in accordance with MIL-N-7513 (3.5.1.1.2).

3.5.1.2 Top and bottom plastic sheets.-The top and bottom plastic sheets shall be cut to the dimensions given in figure 4, 5, or 6, whichever is applicable (3.5.4). The bottom sheet shall have a nominal 3/32 inch thickness. The top sheet shall have a nominal 0.010 inch thickness and shall be marked as specified in 3.5.1.1.1.

3.5.1.3 Paper insert.-The paper insert shall be cut to provide an opaque backing for the central and lower scales for each type of computer. It shall be positioned as shown by the dotted lines in figure 4, 5, or 6, whichever is applicable. The paper shall not come closer than 3/32 inch to any edge of the plastic and it shall not cover any portion of the upper scale. The upper scale is nearest the swivel. The paper may back up any number, letter, or word associated with but not contained in the central and lower scales proper; however, no number, letter, or word shall be only partially backed up by the paper. The paper insert shall be marked as specified in 3.5.1.1.2.

3.5.1.4 Bonding.-The printed paper insert shall be bonded between the marked top plastic sheet and the bottom plastic sheet. Bonding shall be complete and permanent without

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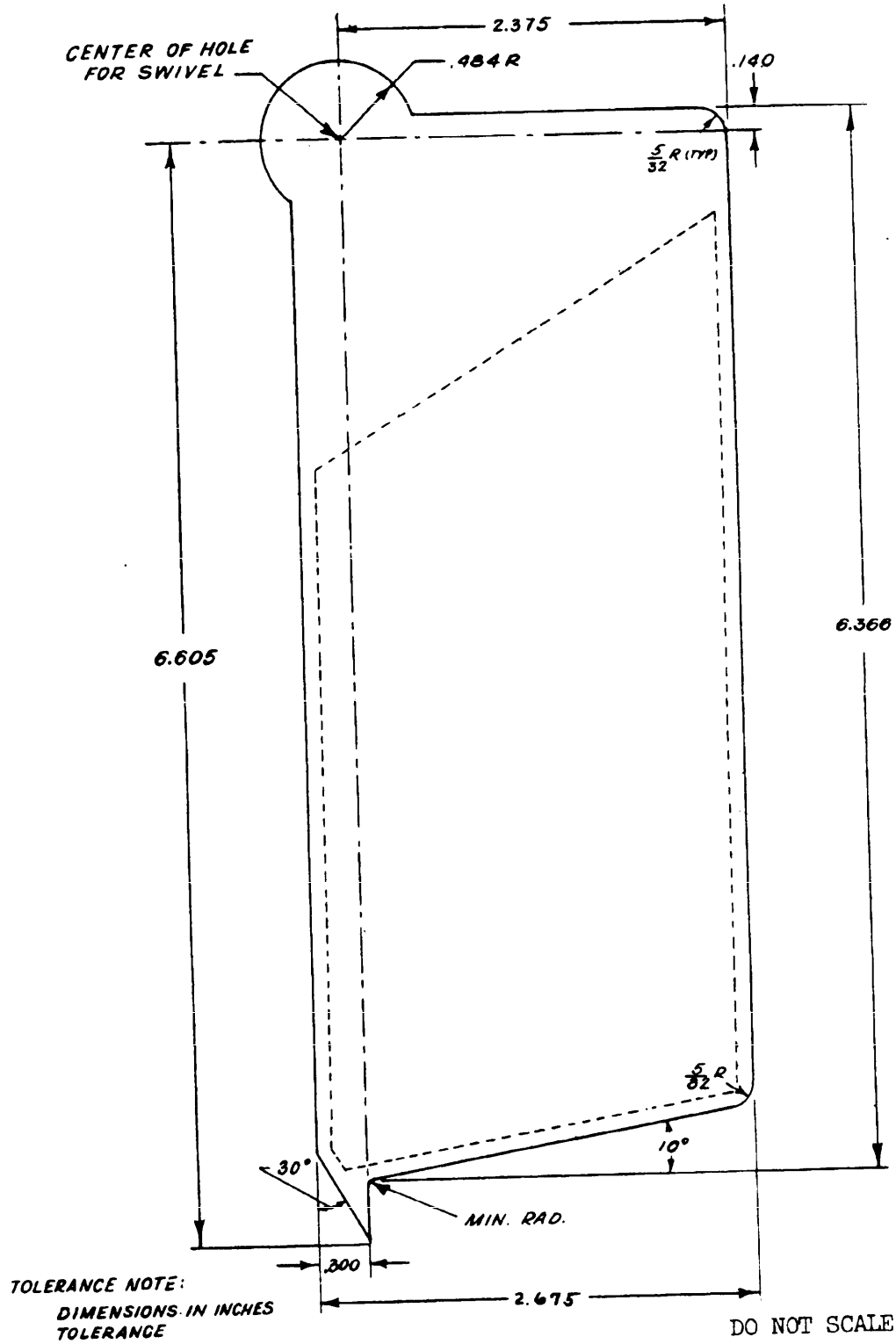


FIGURE 4 BODY (ML-543/UM)

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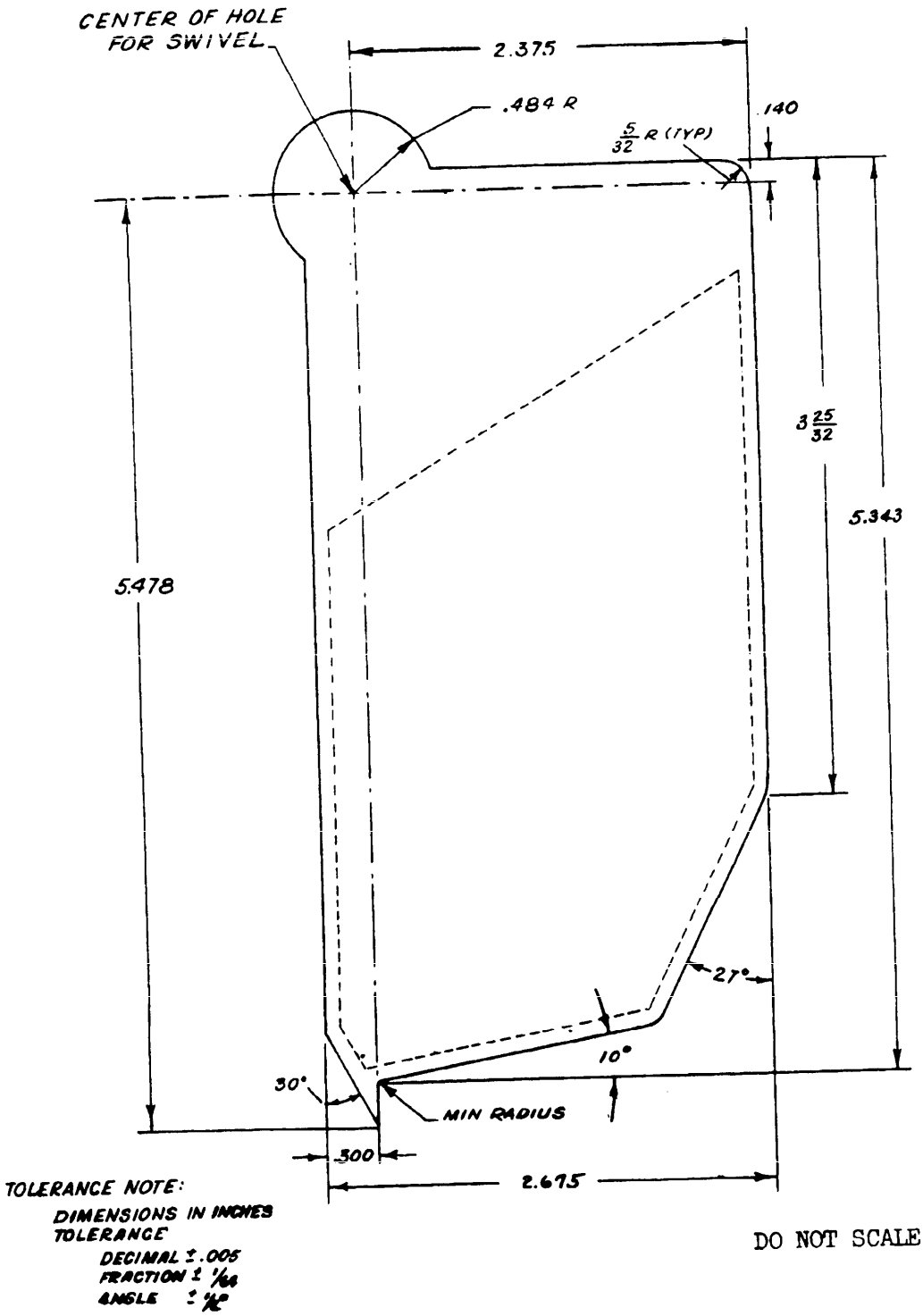


FIGURE 5 BODY(ML-544/UM)

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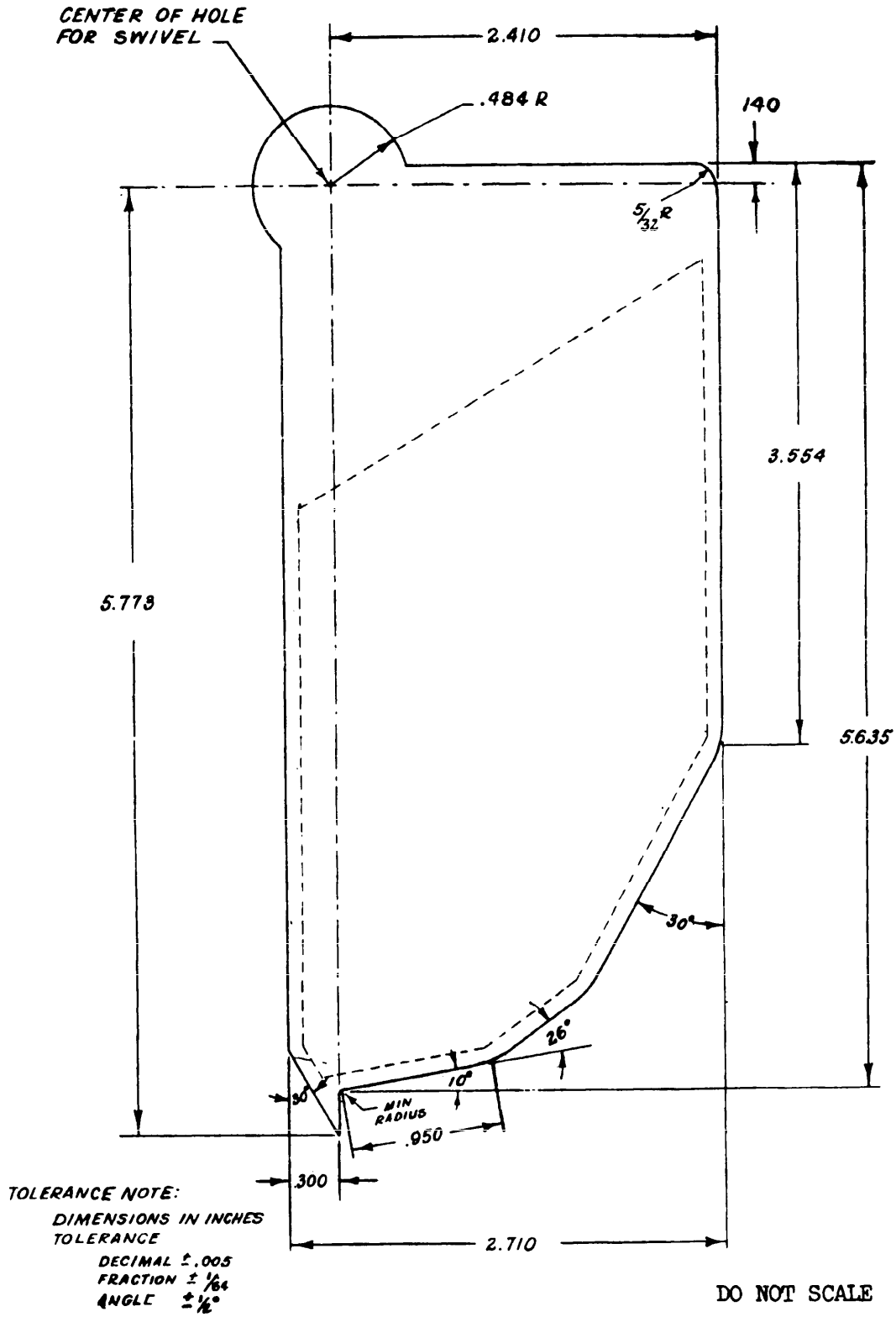


FIGURE 6 BODY(ML-545/UM)

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noticeable air inclusions. There shall be no distortion of the markings or loss of plastic sheet transparency as a result of the bonding.

3.5.2 Cursor.-The cursor shall be cut to the dimensions of figure 7, 8, or 9, whichever is applicable (3.5.4). It shall have a nominal 3/32 inch thickness. The etched line shown in the figure shall be on the bottom (top view is shown in figure). It shall be 0.010 inch wide, plus or minus 0.002 inch, and shall be colored black.

3.5.3 Swivel.-The design of the swivel which attaches the cursor to the main body shall be such that the following requirements are met:

3.5.3.1 Relative motion.-When the pointers of the computer are coincident, the torque necessary to cause impending motion in the normal direction of movement shall be 25 plus or minus 15 inch-ounces. The swivel shall permit only angular motion between the body and the cursor. Translational motion shall not exceed 0.002 inch.

3.5.3.1.1 Change of swivel characteristics.-After rotating the cursor 360 degrees clockwise and 360 degrees counterclockwise for one hundred 360 degree turns each, the swivel shall not loosen or tighten enough so that it fails to meet requirement 3.5.3.1.

3.5.3.2 Size limitations.-The swivel assembly shall not extend above the cursor more than 3/16 inch or below the body at all.

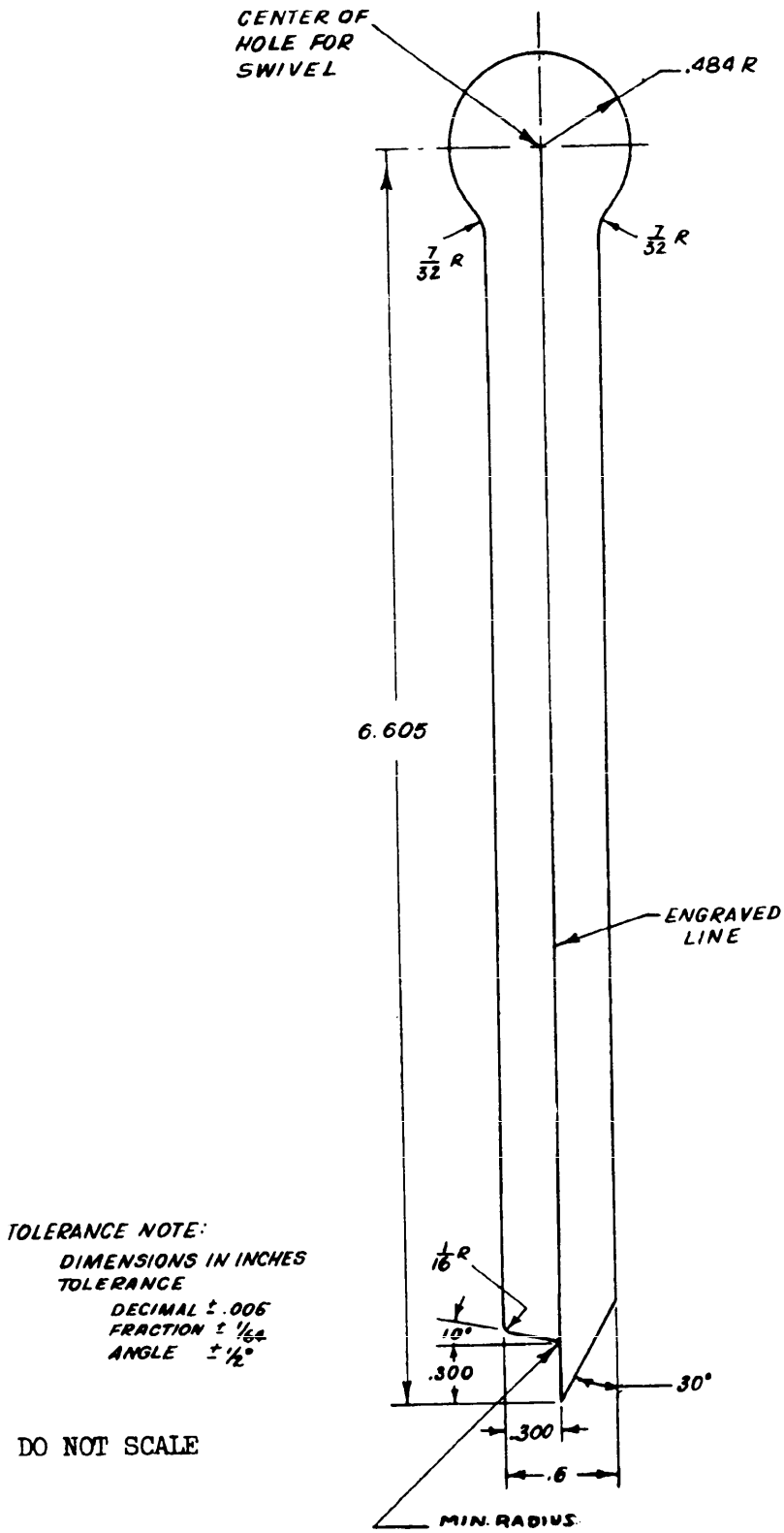
3.5.3.3 Other swivel considerations.-The swivel assembly shall not permit disassembly and shall require no adjustments after leaving the manufacturer. It shall not require periodic lubrication, and any lubrication required shall be accomplished by the manufacturer at the time of assembly.

3.5.4 Applicable figures:

Computer	Applicable figures
ML-543/UM	1, 4, 7
ML-544/UM	2, 5, 8
ML-545/UM	3, 6, 9

3.6 Workmanship.-The computers, including all parts, shall be fabricated and finished in a workmanlike manner. Particular attention shall be given to assure quality of the bonding and to maintain the high polish finish of the plastic sheets.

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TOLERANCE NOTE:
DIMENSIONS IN INCHES
TOLERANCE
DECIMAL ± .005
FRACTION ± 1/64
ANGLE ± 1/2°

DO NOT SCALE

FIGURE 7 CURSOR (ML-543/UM)

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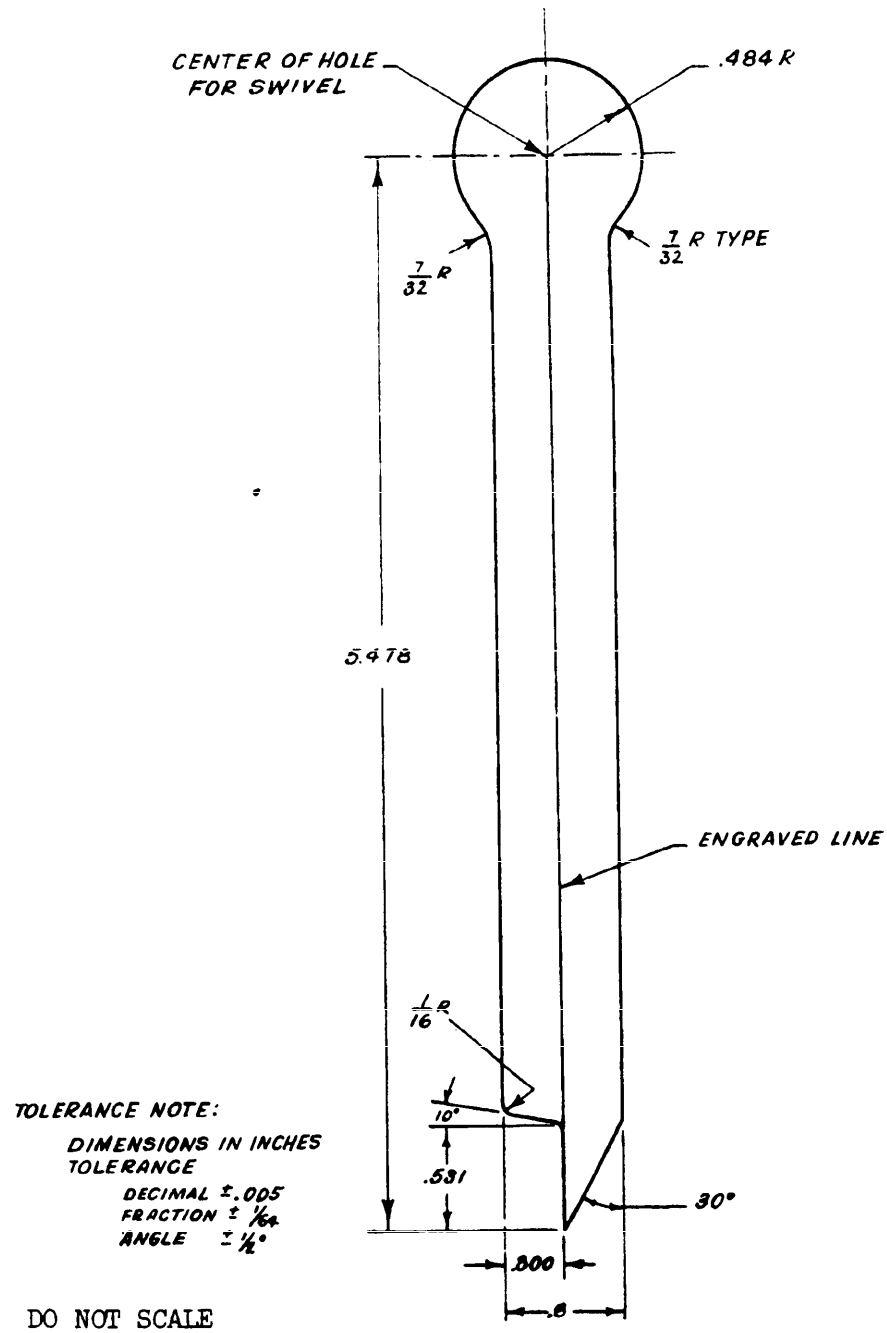


FIGURE 8 CURSOR (ML-544/UM)

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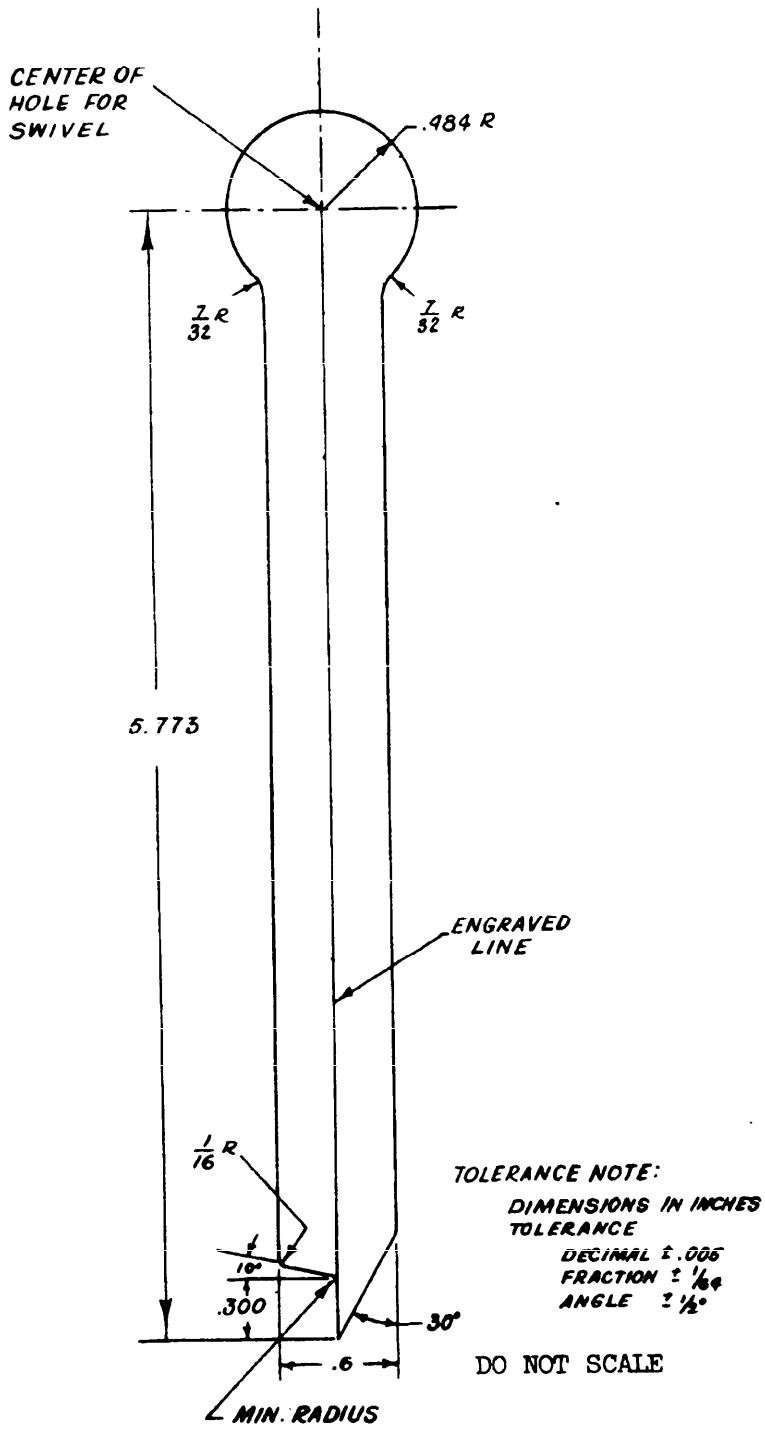


FIGURE 9 CURSOR (ML-545/UM)

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3.6.1 Finish.-The finished computers shall be free from burrs and rough or sharp edges of metal and plastic parts alike, except for the points on the cursor and the body. All edges and corners shall be rounded, except for the points on the cursor and body. The surfaces shall be free from cracks, seams, bubbles, or raised spots.

3.6.2 Flatness.-There shall be no evidence of waves or ripples in the surface of the computers when viewed at an angle of 30 degrees to the line of sight of the observer. The total deviation from a flat surface shall not exceed 1/32 inch either in width or length.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Classification of tests.-The inspection and testing of the computers shall be classified as follows:

a.	Acceptance tests	(4.3)
b.	Preproduction tests	(4.6)

4.3 Acceptance tests.-Acceptance tests shall consist of:

a.	Individual test	(4.3.1)
b.	Sampling plan and tests	(4.3.2)

4.3.1 Individual test.-Each computer shall be subjected to the following test as described under 4.5, Test methods, of this specification: Examination of product (4.5.1).

4.3.2 Sampling plan and tests.-Five computers shall be selected at random from each production run of 500, or fraction thereof produced on the order, and subjected to all the tests described under 4.5, Test methods.

4.3.2.1 Rejection and retest.-When one item selected from a production run fails to meet the specification, no items on hand or later produced shall be accepted until the extent and cause of the failure are determined. After corrections have been made, all necessary tests shall be repeated.

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4.3.2.2 Individual tests may continue.-For operational reasons, individual tests may be continued pending investigation of a sampling test failure. But final acceptance of items on hand, or produced later, shall not be made until it has been determined that items meet all requirements of the specification.

4.3.3 Defects in items already accepted. - The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.4 Test conditions

4.4.1 Atmospheric conditions.-Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of 28 to 32 inches of mercury and at a temperature of 25 degrees C, plus or minus 10 degrees C, and a relative humidity of 80 percent or less. Where tests are made with atmospheric pressure or temperature substantially different from the above values, proper allowances shall be made for the changes in instrument reading.

4.5 Test methods

4.5.1 Examination of product.-Each computer shall be inspected to determine compliance with this specification with respect to material, dimensions, paper shrinkage, markings, workmanship, printing plate damage, conformance to figures, bonding, flatness, and finish.

4.5.2 Heat and humidity test.-The computers shall be exposed to a temperature of plus 54 degrees C at a relative humidity of 90 plus or minus 5 percent for 24 hours. After the exposure, the computers shall be placed on a flat surface. When in this position, no portion of the surface shall be more than 1/16 inch above the surface on which it is resting. In addition, there shall be no sign of discoloration or separation of the laminations.

4.5.3 Low temperature test.-The computers shall be exposed to a temperature of minus 40 degrees C at any relative humidity for 8 hours. After the exposure, the computers shall be placed on a flat surface. When in this position, no portion of the surface shall be more than 1/16 inch above the surface on which it is resting. In addition, there shall be no sign of discoloration or separation of the laminations.

4.5.4 Performance test.-Each computer selected for performance testing shall be subjected to the following test.

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4.5.4.1 The cursor shall be rotated 360 degrees clockwise and 360 degrees counterclockwise for one hundred 360 degree turns each. At the end of this test, the torque necessary to turn the swivel shall not be changed by more than 10 percent of its value at the beginning of the test. Also, translational motion between the cursor and the body shall not exceed 0.002 inch.

4.6 Preproduction tests

4.6.1 Preproduction test samples.-The preproduction test samples shall consist of two models representative of the production equipment. They shall be tested at a laboratory designated by the procuring activity or, when so stated in the contract, at the contractor's plant under the supervision of the procuring activity.

4.6.2 Preproduction tests.-Preproduction tests shall consist of all tests described under 4.5, Test methods.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing.-Preparation for delivery shall be in accordance with MIL-P-7936. Preservation for level A shall be in accordance with MIL-P-116, method IA.

5.2 Marking for shipment.-Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. The shipment marking nomenclature shall be as follows: Computer, Wind Geostrophic and Gradient ML- (use appropriate number).

6. NOTES

6.1 Use.-The computers described in this specification are intended for use by weather personnel in the determination of geostrophic and gradient winds.

6.2 Ordering data.-Procurement documents should specify the following:

a. Title, number, and date of specification.
b. Level of preservation, packaging, and packing desired.
c. Type and quantity of computer desired.
d. Stock number of each computer on order for inclusion in the identification marking.
e. The availability of inspection records specified in 4.1.
f. Point of preproduction testing.
g. Preproduction samples.-It is expected that the contract or purchase order will specify that two computers of each type ordered will be required as preproduction samples and that these samples will be subjected to the preproduction tests to determine

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compliance with the requirements of this specification. The invitation for bids and the contract should specify the point of inspection for these tests.

6.3 Exact double scale masters of the information contained in figures 1, 2, and 3 are on file at the Air Force Cambridge Research Center, ATTN: CRZWB, and will be made available to the manufacturer. These can be used with photographic or other suitable reproduction processes.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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Navy - Bureau of Aeronautics
Air Force

Preparing Activity:
Air Force

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